



Understanding AI in 2026

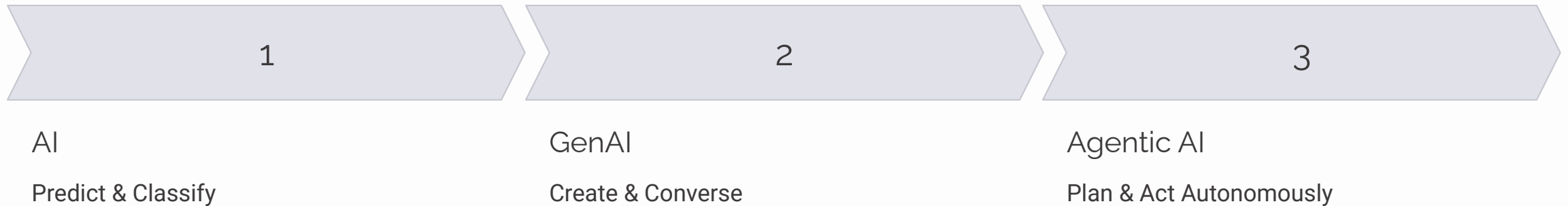
From classic algorithms to autonomous agents — a practical mental model for navigating the AI landscape, understanding the economics of large language models, and choosing the right tools on Amazon Bedrock.

MENTAL MODELS · ECONOMICS · AMAZON BEDROCK

The Big Picture

AI, GenAI, and Agentic AI — What's the Difference?

Think of these three as a ladder of increasing capability and autonomy. Each builds on the one before it — but they solve very different problems. Getting this mental model right helps you pick the right tool for the job.



Each tier unlocks new possibilities — but also introduces new complexity, cost, and risk. Understanding where your use case sits is the first step to building effectively.

Traditional AI — The Rule-Follower

What it does: Learns patterns from labeled data to make predictions or classifications. It does one thing, and it does it well.

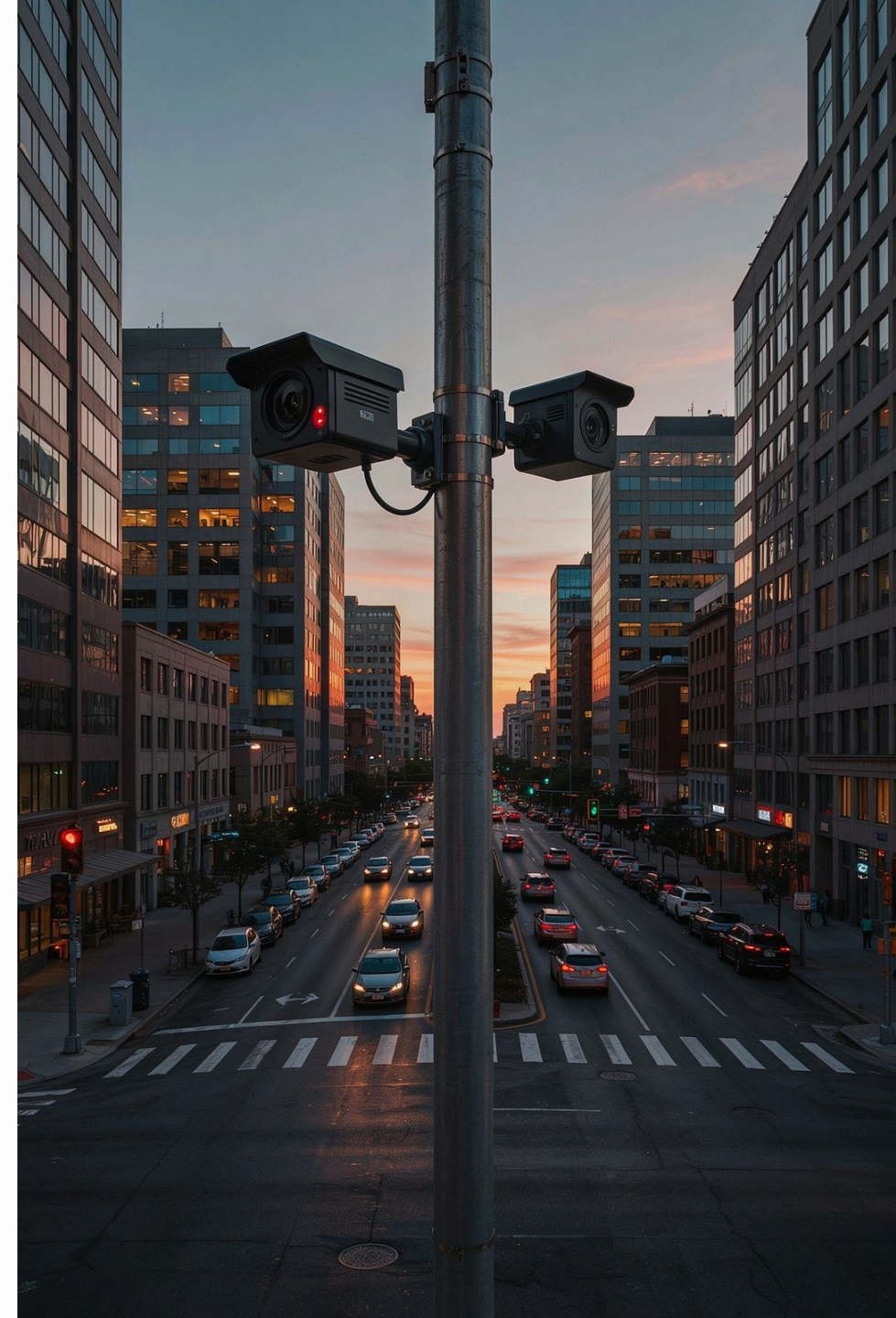
Real-World Example

A traffic camera that detects whether a car is speeding. It was trained on millions of images, and now it confidently classifies: *speeding* or *not speeding*. It cannot write a report, hold a conversation, or decide what to do next.

Other Examples

Spam filters in your email inbox. Netflix recommending your next show. A bank flagging a fraudulent transaction. All narrow, all highly tuned, all brilliant at their single task.

 **Key mental model:** Traditional AI = a specialist with one job. Fast, efficient, but inflexible.



Generative AI — The Creative Collaborator

GenAI doesn't just classify — it **creates**. It was trained on vast amounts of text, code, and images, enabling it to generate new content, summarize documents, write code, and hold natural conversations.



Customer Support Bot

Instead of picking from a fixed menu of answers, a GenAI bot reads your question and writes a tailored response — like a human agent would.



Contract Drafting

A lawyer asks: "*Draft an NDA for a SaaS vendor.*" GenAI produces a full draft in seconds — saving hours of work.



Code Generation

A developer describes a function in plain English. GenAI writes the Python code, adds error handling, and even suggests unit tests.

📌 **Key mental model:** GenAI = a brilliant intern. It creates on demand, but still needs you to review, direct, and approve every output.

Agentic AI — The Autonomous Operator

Agentic AI doesn't just respond to a prompt — it **pursues a goal**. It can plan multi-step tasks, use tools (like web search, databases, or APIs), make decisions, and loop until the job is done.

The Goal

"Book me the cheapest flight to New York next Tuesday, add it to my calendar, and email my team the itinerary."

A human would do this in 10 steps across 4 apps. An Agentic AI does it autonomously — no hand-holding required.

How it works

- 1** Perceive
Understands the goal and context
- 2** Plan
Breaks it into sub-tasks
- 3** Act
Calls tools, APIs, and services
- 4** Iterate
Checks results, adjusts, and completes



Mental Model Summary

Three Tiers at a Glance

Use this table as a quick reference whenever you're deciding which type of AI fits your use case.

Dimension	Traditional AI	Generative AI	Agentic AI
Core Skill	Predict / Classify	Create / Converse	Plan / Act
Human Analogy	Specialist employee	Brilliant intern	Independent contractor
Real Example	Fraud detection	ChatGPT, Claude	Auto-booking travel
Needs Supervision?	Low (after training)	High (review outputs)	Medium (set guardrails)
Best For	Structured, repetitive tasks	Open-ended content & Q&A	Multi-step workflows

LLM Economics — What Does It Actually Cost?

Running a large language model isn't free. Understanding the cost structure helps you build efficiently and avoid surprise bills. LLM pricing typically breaks down into three key dimensions.

Tokens

You pay per token — roughly $\frac{3}{4}$ **of a word**.
Input tokens (your prompt) + output tokens (the model's response) = your bill. A 1,000-word essay is ~1,300 tokens.

Model Size

Bigger models (more parameters) are smarter but cost more per token. A frontier model like Claude 3.5 Sonnet costs ~10–50x more than a small/lightweight model like Claude Haiku.

Context Window

The longer the conversation or document you feed in, the more tokens consumed. A 100-page PDF loaded into context can cost as much as hundreds of short queries.

LLM Economics — Choosing the Right Model for the Job

The golden rule: **don't use a sledgehammer to crack a nut**. Matching model capability to task complexity is the single biggest lever for cost optimization.



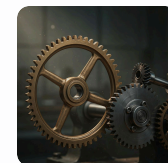
Simple Tasks → Small Models

Classifying a support ticket as "billing" or "technical"?
Summarizing a 3-sentence update? Use a fast, cheap model.
Cost: fractions of a cent per call.
Example: Claude Haiku, Titan Lite.



Complex Tasks → Frontier Models

Writing a legal brief, generating production-ready code, or analyzing a 50-page earnings report? Invest in a frontier model for accuracy that matters.
Example: Claude 3.5 Sonnet, Llama 3.3 70B.



Agentic Workflows → Orchestrate Wisely

In a multi-step agent, use a powerful model for reasoning and planning, but route simple sub-tasks (like formatting output or looking up a fact) to cheaper models to keep costs under control.

Amazon Bedrock — One Platform, Many Model Families

Amazon Bedrock is AWS's fully managed service for accessing foundation models via API — no infrastructure to manage. It offers a curated marketplace of leading model families, letting you swap models without changing your application architecture.



Anthropic — Claude

Best-in-class reasoning, safety, and long context. Claude 3 family (Haiku, Sonnet, Opus) covers the full cost/capability spectrum. Ideal for enterprise Q&A, document analysis, and agentic tasks.



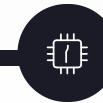
Meta — Llama 3

Open-weight models offering strong performance at lower cost. Llama 3.3 70B rivals frontier models for many tasks. Great for organizations wanting flexibility and fine-tuning control.



Amazon — Titan & Nova

AWS's own model family. Titan covers text embeddings and generation. Nova (launched 2024) brings multimodal capabilities. Deeply integrated with AWS services like S3, Kendra, and OpenSearch.



Mistral, Cohere & More

Mistral excels at multilingual and code tasks. Cohere specializes in enterprise search and RAG (Retrieval-Augmented Generation). Stability AI powers image generation use cases.

Key Takeaways

Your AI Decision Framework

Whether you're an engineer, product manager, or executive — these three questions will guide every AI decision you make.

1 What kind of intelligence do I need?

Classify → Traditional AI.

Create/converse → GenAI. Multi-step autonomous tasks → Agentic AI. Start here before picking any tool or vendor.

2 What is the cost/quality tradeoff?

Match model size to task complexity.

Use small, fast models for routine work. Reserve frontier models for high-stakes, nuanced outputs. Architect agentic workflows to route intelligently.

3 Which Bedrock model family fits?

Claude for reasoning and safety. Llama for flexibility and fine-tuning. Titan/Nova for deep AWS integration. Cohere for enterprise search. Mix and match — Bedrock makes it easy to switch.

 **Remember:** The best AI solution is the simplest one that solves the problem reliably, at a cost that makes business sense.